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RESEARCH ARTICLE

# Lost miRNA surveillance of Notch, IGFR pathway—road to sarcomagenesis

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**Abstract** The goal of this study was to compare and analyze differentially expressed miRNA and their targets in human chondrosarcoma JJ012 and chondrocytes C 28 cell lines (control) to elucidate deregulation of major signal transduction pathways involved in sarcomagenesis. Total RNA extraction was followed by analyzing RNA quality and integrity. Exiqon human miRNA panel of 743 unique miRNA assays and Illumina microarray HT-12 platform and quantitative reverse transcriptase–PCR verification of targets were performed. The results from human miRNA Exiqon arrays with biological triplicates indicated 28 significant miRNAs ($P$ value ≤0.01). A total 3,045 target genes were derived from the miRWalk database for these 28 miRNAs with 587 common and 2,458 unique target genes. The results of our analyses of the significantly downregulated and upregulated miRNAs in chondrosarcoma cell line indicated the predominant dysregulation of NOTCH, insulin-like growth factor receptor (IGFR), and downstream rat sarcoma, and Src pathways, compared to control. Among the upregulated targets for upregulated miRNAs were the cluster of cancer testis antigen (CTA) genes, located on X chromosome, and their expression was correlated to IGFR pathway activity. Based on our observations, lost miRNA surveillance of NOTCH and IGFR pathways is involved in and leads to sarcomagenesis. We conclude that upregulation of CTA genes is due to hypomethylation that are controlled by epi-miRNAs. We do not preclude the possibility that the upregulated miRNAs, which target CTA genes located in adjacent regions in chromosome X, are epi-miRNAs that influence target gene expression by directly regulating epigenetic processes. Future endeavors will be directed towards understanding the posttranscriptional modifications that affect miRNA expression in sarcomas.

**Keywords** Notch · IGFR · Sarcomagenesis · Chondrosarcoma · miRNA · Chondrocytes

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## Introduction

Sarcomas are rare malignant tumors that are derived from mesenchymal cells. The signaling events resulting in mesenchymal cell transformation to sarcoma are not known. Some tumor subtypes, especially chondrosarcoma, do not respond to chemotherapy or radiotherapy; thus, surgery is the only treatment option [1–5].

Increased insulin-like growth factor receptor (IGF1R) levels or some of its ligands and dysregulation of IGF axis have been correlated with sarcomagenesis [6–16]. IGF1R is overexpressed in both bone and soft tissue sarcomas and is associated with aggressive behavior in these tumors. Clinical evidence suggests that benign cartilage lesions can progress to chondrosarcoma, and the IGFR has been implicated in this

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process [17]. The insulin-like growth factor receptor contains two intracellular beta subunits with tyrosine kinase domains responsible for activating the following: insulin receptor substrate (IRS), phosphatidylinositol 3 kinase (PI3K), protein kinase B (AKT), mammalian target of rapamycin (mTOR), and rat sarcoma (RAS)/RAF/mitogen-activated protein kinase (MAPK) pathways that promote cell growth and cancer cell transformation [18]. The upstream of IGFR Notch-1 stimulates survival of cancer cells during hypoxia by activating the IGF1R pathway and IGF1R expression by directly regulating its promoter. Aberrant activation of the Notch signaling pathway is implicated in various neoplastic processes, contributing to aggressiveness for sarcomas, although limited information regarding Notch deregulation in chondrosarcoma is currently available [18–22]. Altered Notch status is associated with both pro- and antitumor suppressive roles. The effects of Notch signaling, involved in cell–cell communication, in tumor behavior are dependent on cellular context. The involvement of other pathways such as Hedgehog and Wnt in sarcomagenesis has been reported, although Wnt pathway involvement is controversial [23].

The sentinel work on *Caenorhabditis elegans* by V. Ambros and colleagues in 1993 on miRNA changed the approach and our view on the cellular events and molecular basis of malignant transformation [24]. Studying on gene damage is not enough; we must now investigate the processes that alter miRNA. This will lead to a fundamental understanding on why cancer seems to occur randomly, and the ways it can be controlled or prevented. With the discovery of noncoding RNA (ncRNA) transcripts with no significant open reading frame, it has become evident that the genomic complexity of the cancer cell is far greater than expected [24]. Examination of tumor-specific microRNA expression profiles has revealed widespread deregulation of these molecules in diverse cancers. Several experiments suggest that microRNAs may function as a novel class of oncogenes and tumor suppressor genes [25, 26]. MicroRNAs exert their functions through binding in the 3′ UTR of genes and regulation of their expression. MicroRNAs can bind with perfect or imperfect complementarity to each strand of the double-stranded RNA and, depending on that, regulate gene expression. There are three major known mechanisms for miRNA-mediated gene regulation: translation repression, direct mRNA degradation, and miRNA-mediated mRNA decay. Which mechanism controls gene expression is entirely dependent on the degree of miRNA complementarity to their targeted mRNAs. However, there is growing evidence that small RNAs can also serve as activators of gene expression by targeting genes with complementary promoter sequences. This novel mechanism, known as RNA activation, appears to be conserved in at least mammalian cells and triggered by both endogenous and artificially designed small RNAs [27, 28]. Interestingly, more than 50 % of miRNA genes can be found in cancer-associated genomic regions or in fragile sites, suggesting that miRNAs play an important role in the pathogenesis of neoplasias. They are located also in minimal regions of heterozygosity loss, minimal regions of amplification, or common breakpoint regions that are genetically altered in human cancers [29]. Transcription factors are subject to miRNA regulation and miRNA cell-type specificity, and unlike transcription factors, miRNAs can rapidly turn off protein production right at the site of protein production, the ribosome. Finally, miRNAs are involved in metabolic reprogramming to oncogenesis [30]. Malignant solid tumors represent a heterogeneous group of tumors which compose a significant proportion of all cancer cases annually. Sarcomas are malignant tumors of mesenchymal (connective tissue) origin, which comprise 10–15 % of all childhood tumors and proportionally less in adults. However, relative to other malignant solid tumors, sarcomas continue to be associated with a disproportionate percentage of morbidity and mortality. Genetic and epigenetic deregulation occurring at defined points during the developmental program of progenitor cells can give rise to malignant tumors of bone (osteosarcoma or Ewing's sarcoma), striated muscle (rhabdomyosarcoma), smooth muscle, uterine (leiomyosarcoma), fat (liposarcoma), cartilage (chondrosarcoma), or fibrous tissue (fibrosarcoma)—all originating from connective tissue structures derived from mesenchymal cells. The insulin-like growth factor pathway represents one key signaling pathway intimately involved in Ewing sarcoma sarcomagenesis. EWS/FLI-1 chromosomal translocation has previously been shown to directly target and regulate IGF1 and the IGF1 receptor (IGF1R). These observations describe an emerging biological principle, where the products of translocation of fusion genes with appropriate cellular microenvironment exert their oncogenic properties by deregulating tumor-specific miRNAs via direct interaction with miRNA regulatory sequences. For example, reporter gene assays utilizing putative miR 145 promoter demonstrated that reporter expression was decreasing upon transient expression of EWS/FLI-, which suggested direct interaction of miRNA and fusion genes in sarcoma [31].

The current study presents miRNA and targets comparative analysis of human chondrosarcoma JJ012 versus control chondrocytes C 28 cell lines, elucidating the roles of miRNA in major signal transduction pathways leading to sarcomagenesis.

## Methods

*Tissue culture* Chondrosarcoma JJ012 and C28 cells were cultured in monolayers as previously described [3, 4].

*Cell lysis with QIAzol, RNA extraction, and quality control* Cells were seeded on a 60×15-mm Petri dish at the density of $5\times10^6$ and were incubated 24 h. The cells were

lysed with 700 μl of Qiazol per dish. One hundred micrograms of total RNA from the sample was prepared (Qiagen RNeasy mini kit).

*MirCURY LNA Universal RT microRNA human Panel I+ II (version 2.0)* RNA quality was checked with a NanoDrop 8000 Spectrophotometer for RNA integrity. The presence of small RNA fractions was determined using a Bioanalyzer 2100 (Agilent, Santa Clara). Twenty nanograms of total RNA per panel was reverse transcribed using the miRCURY LNA™ Universal RT microRNA PCR and Polyadenylation and cDNA synthesis kit (Exiqon, Woburn, MA). For each sample, the cDNA was diluted, and each reaction was combined with SYBR Green Master Mix (Exiqon) and added to the Ready-to-Use PCR panels. Human microRNA Ready-to-Use PCR panels I and II hold 743 different miRNA targets and six reference gene assays (mirBase13). They were tested on a Roche LightCycler 480 real-time PCR instrument. The data was analyzed using MultiD GenEx (MultiD, Gothenburg, Sweden), and the raw data was inter-plate calibrated; outliers and false positives were filtered. All missing values were imputed, and data was normalized to the global mean. Fold change and *P* values (Benjamini–Hochberg) were calculated following relative quantification and log base2 conversion.

*Microarray analysis and statistics* Expression data was obtained by hybridizing samples to HumanHT-12 V4_0_R1 array platform (Illumina) that were loaded on GeneSpring® 12.5 GX (Agilent Technologies, Santa Clara, CA, USA). The raw intensities from the array were normalized using the "quantile normalization" method and were log (base 2)-transformed. The normalized data was subjected to statistical analysis using paired Student's *T* test (two-class comparisons). All significant and differentially expressed (DE) genes were subset using a *P* value ≤0.005 and fold change ≥2 thresholds. *P* values were corrected for multiple testing using the Benjamini–Hochberg (BH) method.

*qRT-PCR* After extraction, the total RNA was treated with DNase (Ambion) to remove genomic DNA contamination. Sample concentrations were normalized to 133.3 ng/μl and 2,000 ng were used to generate cDNA with the iScript Advanced cDNA synthesis kit (Biorad, Carlsbad CA). The PrimePCR custom plates (384-well format, Biorad) contained 20× stock lyophilized primers for one gene per well and were optimized for use with SsoAdvanced SYBR Green supermix from Biorad. Ten microliters of the mix was transferred (diluted cDNA plus SsoAdvanced SYBR Green supermix) into each well on the 384-well plate. Real-time PCR was performed on the "Biorad CFX384 Real-Time PCR detection system," and data analyses were done using CFX Manager 3.0 software.

*miRNA–mRNA expression* All significant miRNAs (*P* value (BH corrected) ≤0.01) were searched against the miRWalk database (http://www.umm.uni-heidelberg.de/apps/zmf/mirwalk/index.html) [32] on predicted and validated microRNA targets. Gene expression data for all the significant (*P* value ≤0.05) validated or predicted gene targets for each miRNA was extracted from the Illumina gene expression microarray and matched against the miRNA expression.

## Results

### Statistically significant differentially expressed miRNAs in human JJ012 chondrosarcoma and human chondrocytes C28 (control) cell lines

We have performed miRNA and target comparative analysis of human chondrosarcoma JJ012 versus control chondrocytes C 28 cell lines. Cell culture, lysis, RNA extraction and quality control, and isolation of total RNA, followed by reverse transcription using the miRCURY LNA™ Universal RT microRNA PCR and Polyadenylation and cDNA synthesis kit were performed as described in the "Methods" section. To achieve the highest stringency and accuracy, we have narrowed the statistical parameters (*P* value ≤0.01 and fold change ≥2). The results from human miRNA Exiqon arrays with biological triplicates indicated 28 significant miRNAs (*P* value ≤0.01) among the 188 downregulated and 148 upregulated miRNAs when comparing their expression in a chondrosarcoma cell line to that of control C28 chondrocytes. The miRNA microarray database is MIAME-compliant and has been deposited in the GEO database http://www.ncbi.nlm.nih.gov/geo (submission record GSE48420). The expression of 28 statistically significant (10 upregulated and 18 downregulated) miRNAs were confirmed by qRT-PCR (Fig. 1). As it can be seen in the heat map, the upregulated miRNAs are depicted in red zones in descending manner to green zone, corresponding to downregulated genes. We have indicated the fold change upregulation for these ten miRNAs and the number of significant and unique targets correspondingly. The most significantly upregulated ten miRNAs with their significant/unique target genes were miR 551a (73.69 fold) and miR 105 (18.25 fold) (Table 1), whereas among the 18 differentially expressed significantly downregulated human miRNAs, these were miR 886-3p (218.76 fold) and miR 143 (71.83 fold) (Table 2). Most of the upregulated miRNAs as it was anticipated were those with oncogene function, and majority of the downregulated were tumor suppressors, however, with certain exceptions due to epigenetic regulation (let-7) and tissue specificity that is described in more detail in the "Discussion" section.

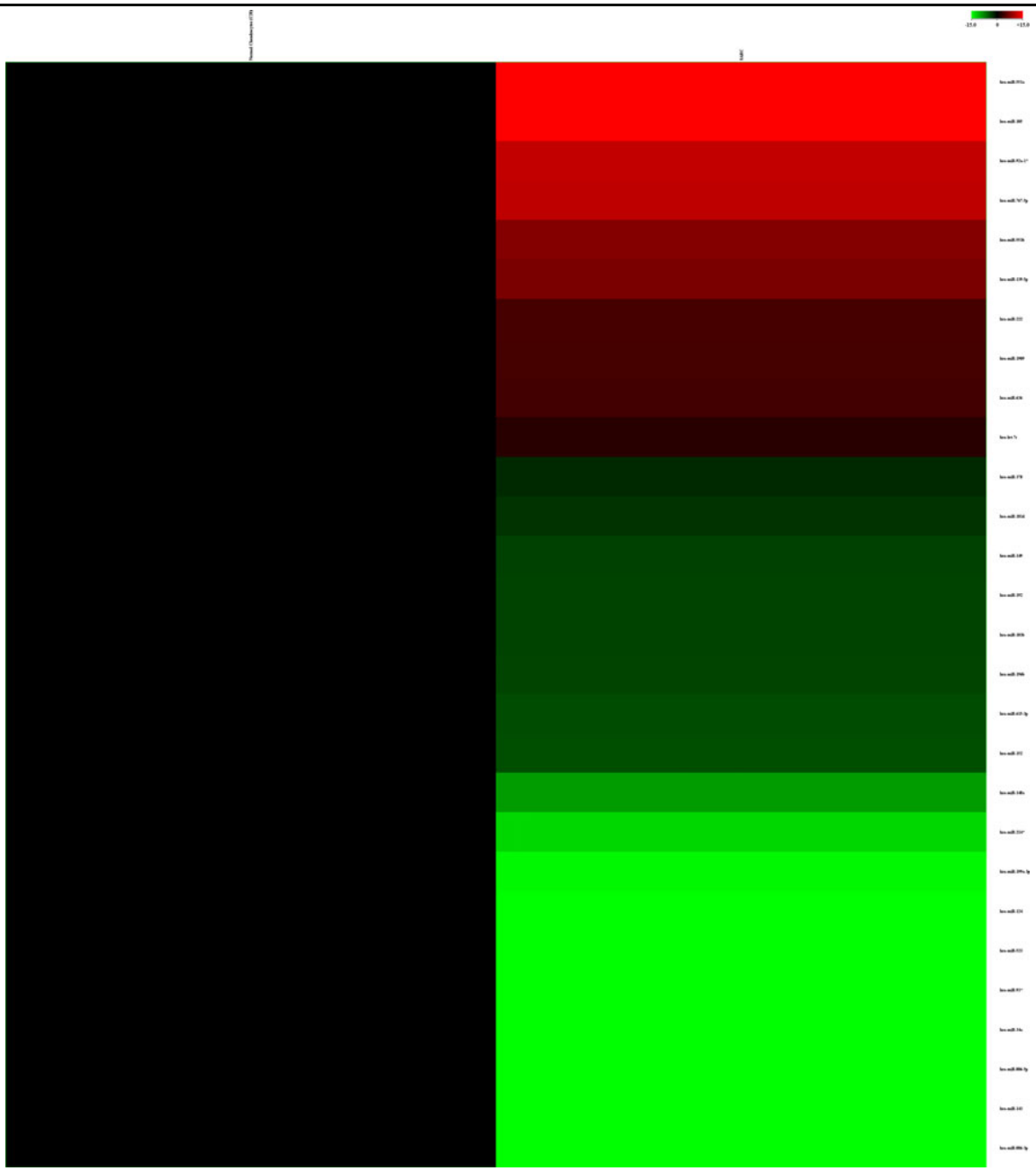


**Fig. 1** Heat map of significant differentially expressed miRNA profile in human JJ012 chondrosarcoma and control C28 chondrocytes cell lines. The upregulated miRNAs are depicted in descending order, from upregulated genes to downregulated genes

Identification of predicted and significant unique target genes, which indicated predominant dysregulation of NOTCH, IGFR pathways

Three-thousand forty five (3,045) target genes in total were derived from the miRWalk database for these 28 miRNAs. Of these 3,045, there were 587 common and 2,458 unique target genes between up- and downregulated miRNAs. The results indicated the predominant dysregulation of NOTCH, IGFR and downstream RAS and Src pathways and their targets (Table 3). Among the predicted targets for miR 886-3p were c Src, stromal-derived factor 1 (SDF1); for miR 143-IGF1R and IGF2R, cSrc; for miR 886-5p CLE16a, Notch 1 and CLE11A (representing Notch family); and for the rest of miRNAS like miRNAs-34A; miR 523; miR 124; miR 199a-3p; miR 214; miR 148a; miR 152; miR 615-3p; miR 196b; miR 181b; miR 192; miR 149; miR 181d; and miR 378, the predominance of RAS family-related oncogenes along with IGFR and NOTCH-related pathway target genes were predicted. Each miRNA displayed targets that are related to that pathway. To confirm the changes in gene expression profile of the target genes (from miRWalk) of these significant

**Table 1** Differentially expressed upregulated miRNAs and their significant/unique target numbers in human JJ012 chondrosarcoma and control C28 chondrocytes cell lines

| miRNA upregulated in human JJ012 chondrosarcoma cell line in comparison with control | Fold change upregulation of miRNA | *P* value (Benjamini–Hochberg) | Significant/unique gene target number ($P \leq 0.01$, fold change ≥2) |
|---|---|---|---|
| miR 551 a | 73.69 | 0.00 | 2 significant, 9 unique |
| miR 105 | 18.25 | 0.00 | 5 significant, 11 unique |
| miR 92a-1 | 11.49 | 0.00 | 28 significant, 63 unique |
| miR 767-5p | 11.15 | 0.00 | 202 significant, 618 unique |
| miR 551b | 7.74 | 0.00 | 1 significant, 4 unique; |
| miR 139-5p | 7.16 | 0.00 | 8 significant, 17 unique |
| miR 222 | 4.13 | 0.00 | 174 significant, 568 unique |
| miR 1909 | 4.07 | 0.01 | 326 significant, 1119 unique |
| miR 636 | 3.95 | 0.01 | 2 significant, 6 unique; |
| miR Let-7c | 2.44 | 0.01 | 127 significant, 521 unique |

**Table 2** Differentially expressed downregulated human miRNAs and their significant/unique target numbers in JJ012 chondrosarcoma and control C28 chondrocytes cell lines

| miRNA (JJ012 human chondrosarcoma cell line) versus control (C28 chondrocytes) | Fold change downregulation of miRNA | *P* value (Benjamini–Hochberg) adjusted values | Significant/unique gene target number ($P \leq 0.01$, fold change ≥2) |
|---|---|---|---|
| miR 886-3p | 218.76 | 0.01 | 1 significant, 3 unique |
| miR 143 | 71.83 | 0.01 | 41 significant, 104 unique |
| miR 886-5p | 70.67 | 0.01 | 1 significant, 5 unique |
| miR 34a | 27.40 | 0.01 | 29 significant, 140 unique |
| miR 93 | 26.78 | 0.01 | 10 significant, 50 unique |
| miR 523 | 25.3 | 0.01 | 11 significant, 39 unique |
| miR 124 | 24.81 | 0.01 | 454 unique, 176 significant |
| miR 199a-3p | 14.58 | 0.01 | 5 significant, 23 unique |
| miR 214 | 12.67 | 0.01 | 8 significant, 34 unique |
| miR 148a | 9.18 | 0.01 | 159 significant, 892 unique |
| miR 152 | 4.65 | 0.01 | 13 significant, 40 unique |
| miR 615-3p | 4.47 | 0.01 | 4 significant, 11 unique |
| miR 196b | 4.02 | 0.01 | 40 unique, 20 significant |
| miR 181b | 3.93 | 0.01 | 49 significant, 130 unique |
| miR 192 | 3.94 | 0.01 | 10 significant, 68 unique |
| miR 149 | 3.76 | 0.01 | 17 significant, 87 unique |
| miR 181d | 2.97 | 0.01 | 28 significant, 63 unique |
| miR 378 | 2.39 | 0.01 | 743 unique, 217 significant |

miRNAs, we performed microarray of these exact same samples on Illumina HT-12 platform. Furthermore, we confirmed with qRT-PCR those significantly expressed top23 up and 23 downregulated target genes (Table 4) that were detected using the microarray. For these series, *P* was chosen as ≤0.05 with fold changes ≥2. Among the most highly upregulated targets were the clusters of cancer testis antigen (CTA) genes, usually upregulated in cancer and testis and residing in X chromosome (Fig. 2), such as CAGE2E (107.88); CAGE2A (101 fold); CAGE8 (95.96 fold); LOC645037 (71.33 fold); and MAGEA 1 (47.68 fold). Among the downregulated targets were PGM (58.38 fold); INPP (17.29 fold); M×1 (6.51 fold); KIAA0363 (6.21 fold); CLDN1 (6.17 fold); and EFEMP1 (5.83 fold).

## Discussion

It is noteworthy that miRNA oncogenes and tumor suppressors should be viewed in cellular context [33, 34]; therefore, it comes as no surprise that certain tumor suppressors manifest themselves as oncogenes in chondrosarcoma (let-7). miR 886-3p, the RNA component of the ribonucleoprotein called vault, was downregulated 218 fold. Among its upregulated targets was cSrc, downstream from the IGFR pathway, and SDF1 also upregulated target and confirmed by Illumina arrays. It was shown in the literature that IGF1 increases the expression of SDF1 receptor CXCR4 and enhances SDF1-induced mesenchymal stem cell migration via PI3-kinase pathway. CXCR4 receptors are essentially involved in SDF1-induced cell mobilization and homing of hematopoietic progenitor cells [35]. Other downregulated miRNAs were miR 143, miR 34a, and miR 124 (Table 2), targeted components of Notch, IGFR, and RAS pathways (Table 3). The upregulated

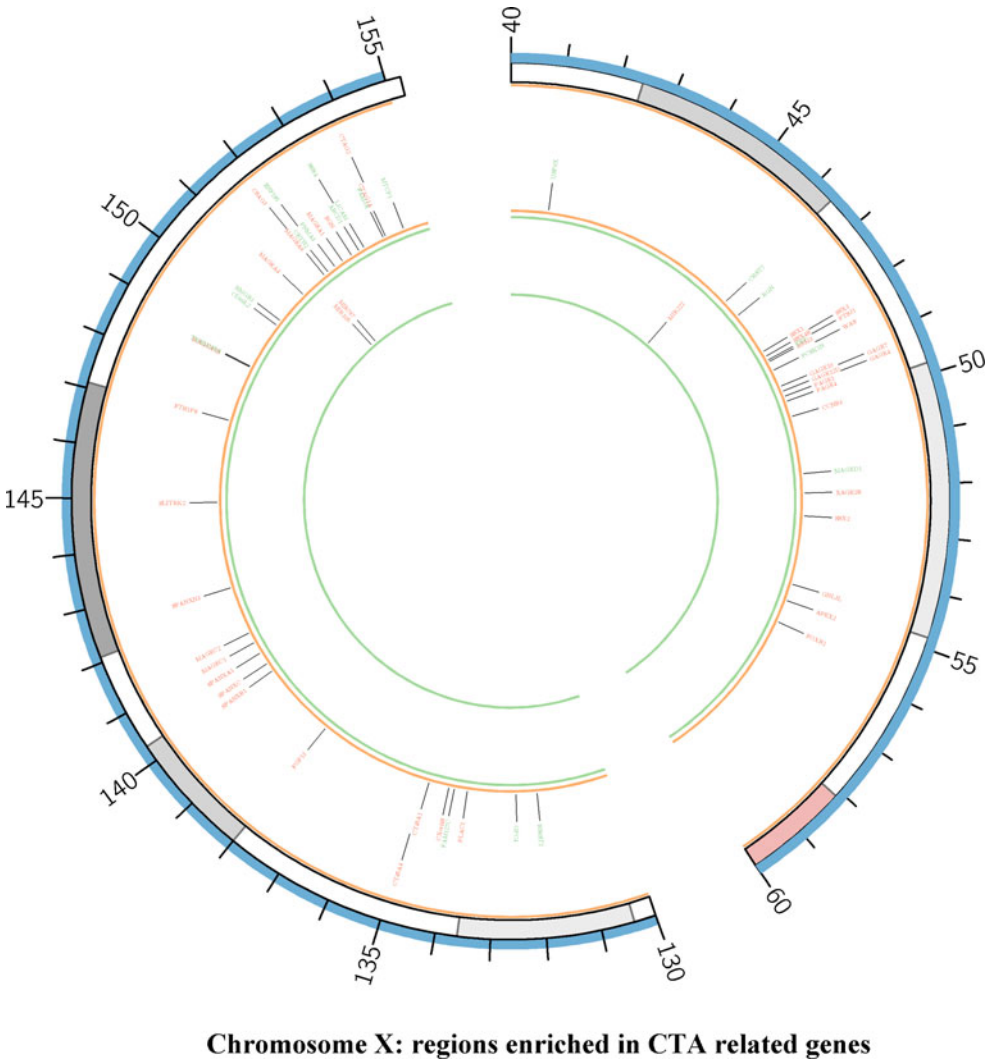



**Fig. 2** Circos plot showing regions of chromosome X enriched for CTA genes. The outer tracks show the genes within the regions, whereas the inner tracks show the miRNA. The list of those genes are displayed in Tables 1 and 2

target for miR 124–caveolin 1 (CAV1), receptor tyrosine kinase, regulates IGF1R internalization and plasma membrane replacement. CAV1 is a target of IGF1R, and its silencing impairs the activation of the IGFR signaling pathway. IGF1R, as receptor tyrosine kinase (RTK) associates with caveolae, invagination of the plasma membrane that regulates vesicular transport endocytosis and intracellular signaling. IGF1R interacts directly with caveolin 1, the most relevant caveolae protein. The fine tunes of these processes can be altered in cancer cell favoring tumor growth/RTK can be internalized via clathrin-coated pit or via caveolae [36]. As we have indicated above, miR 34a was downregulated, and importantly during upregulation, this miRNA directly decreases the protein level of MAGEA, a member of CTA family [37]. Among the upregulated miRNAs were miR 105 and miR 222, let-7 (Table 1). MiR 222, functioning downstream of the RAS pathway, promotes an aggressive cancer phenotype [38]. Let-7 usually acts as a tumor suppressor under Lin-12/Notch pathway regulation [39]. Hypermethylation of let-7a-3 promoter has been described in epithelial ovarian cancer, has paralleled the low expression of insulin-like growth factor II expression, and has been reported to be associated with a good prognosis. Although less frequent, upregulation of certain let-7 family members has also been observed due to hypomethylation [40]. In this study, let-7 was upregulated and lost its tumor suppressor functions, presumably due to hypomethylation, which underscores the importance of the tissue specificity context. Some of let-7 upregulated targets NRAS and STAT activate the RAS-RAF-MAPK pathway, which is downstream from IGFR/EGFR [40]. The targets that were revealed with the Illumina platform confirmed once again the involvement of deregulated Notch and IGFR pathways in sarcomagenesis (Table 3). Among downregulated targets in our chondrosarcoma experiments were claudins (CLDN1), a family of proteins that exhibit tight junctions, which usually act as tumor suppressors [41, 42]. In not pathological conditions [43], the induction of tight junction protein claudin and paracellular permeability during differentiation of osteoblasts

**Table 3** NOTCH/IGFR pathway-related target genes for downregulated miRNA in human JJ012 chondrosarcoma and control C28 chondrocytes cell lines

| miRNA downregulated in human chondrosarcoma JJ012 cell line | Notch/IGFR pathway-related predicted target genes |
|---|---|
| miR 886-3p | SDF-stromal cell derived factor, (CXCL2); cSrc |
| miR 143 | IRS1 (insulin receptor substrate 1); cMyc; ERK (extracellular signal-regulated kinase); KRAS (v-Ki-ras2 Kirsten rat sarcoma viral oncogene homolog); HK2 (hexokinase 2); BRAF (proto-oncogene B-Raf); IGF2R (insulin-like growth factor 2 receptor); IGF1R (insulin-like growth factor 1 receptor); DENN1B (DENN/MADD domain containing 1B) IGFBP5 (insulin-like growth factor binding protein 5); cSrc |
| miR 886-5p | RANBP9 (Ran binding protein 9) (Notch family); IGF1 (insulin-like growth factor 1); RAS GEF (guanine nucleotide exchange factor for Ras-like small GTPases); CLE 16A (CLE protein 16A, Notch family); Notch1; CLE 11A (Notch family); IGFBP5 (insulin-like growth factor binding protein 5); IGF2BP1 (insulin-like growth factor binding protein 10) |
| miR 34a | RAN BP10 (RAN binding protein 10); RAS RAL GDS/AF6 (RAS associated RAL GDS/AF6 domain); RASGRP4 (RAS guanyl releasing protein 4); NOTCH4 |
| miR 93 | TLE (transducin-like enhancer, Notch family); RAB11, RAB5B (RAB 11 and 5B, member s of RAS oncogene family); RAS L12 (Ras-like family 12) |
| miR 523 | SSTR1 (somatostatin receptor 1); EGFR substrate 15 like 1 |
| miR 124 | RAB43; RAB34( RAB43 and RAB 34, members, RAS oncogene family); RANBP10 (Ran binding protein 10); RASGEF A (RAS GEF family member A); RAS assoc/Ral GDS/AF5; RAB 14 (RAB14 protein, RAS oncogene family); GSK3β (Glycogen synthase kinase 3) β; HIFα1 (hypoxia inducible factor); Ras-related GTP binding D; Ras homology gene family member G (Rho G); IGF2BP1 (insulin-like growth factor 2 binding protein 1); PIK3IP1 (phosphoinositide-3-kinase interacting protein 1) |
| miR 199a-3p | IGF1 (insulin like growth factor 1, somatomedin C); MAP3K5 (mitogen-activated protein kinase kinase kinase 5); RAP2A (RAP2A, member of RAS oncogene family) HES; RAPH1 (Ras association (RalGDS/AF-6) and pleckstrin homology domains 1); RREB1 (ras responsive element binding protein 1); PIK3R1 (phosphoinositide-3-kinase, regulatory subunit 1 (alpha)) |
| miR 214 | RAB15 (RAB 15, Ras family oncogene); RAPH1(Ras association (RalGDS/AF-6) and pleckstrin) ; RAB34 (Ras family); TLE4 (transducin-like enhancer of split 4 E (sp1)) |
| miR 148a | DENN/MADD domain; MAP3K4; RASSF6; IGF1; Rab35 (Ras family); RAS protein activating like 1 (RASAL1); PI3KIP1 |
| miR 152 | PREX1 (phosphatidylinositol 3,4,5 Ras exchange); RPTOR (independent companion of mTOR); RICTOR |
| miR 615-3p | RAB37; RAS SF8; RAN (Notch family) TLE (transducin-like enhancer) |
| miR 196b | RANBP10 (RAN binding protein 10) |
| miR 181b | MAP3K2; Lyn (Src family kinase); NOTCH4; KRAS |
| miR 192 | ARHGAP19 (Rho GTPase activating protein 19); IGF1 (somatomedin C); RAB2A (RAB2A, member RAS oncogene family) GLP1R (glucagon-like peptide 1 receptor);v src (vsrc sarcoma virus oncogene) ARHGAP36 (Rho GTPase activating protein 36); PI3KIP1 (Phosphoinositide 3 kinase interacting protein1) |
| miR 149 | RAP2B (Ras-related protein Rap-2b); RAP1A (Ras-related protein Rap-1A) RAP1B (Ras-related protein Rap-1B); IGF2BP1 (insulin growth factor 2 binding protein1); RAB3B; IGFBP5; RAB8B (RAs family) ; Rap 43(Ras family) |
| miR 181d | IRS2 (insulin receptor substrate 2); RALGDS/AK6; ARHGEF3 (Rho guanidine nucleotide exchange factor); RAB11A; KRAS (v-Ki-ras2 Kirsten rat sarcoma viral oncogene homolog) IGF2 (insulin growth factor 2); IGF2BP2 (insulin-like growth factor 2 mRNA binding protein 2) |
| miR 378 | IGF1 (insulin-like growth factor 1); MAPK1 (mitogen-activated protein kinase 1) |

after treatment with IGF1 is regulated via MAP kinase pathway. Furthermore, IGF1 acts as a signal for osteoblast mitogenesis and survival through parallel PI3-kinase and MAP kinase pathways [44], indicating that the signaling pathways absolutely necessary for normal cell function become deranged in malignancy. Retinol binding protein 4,

**Table 4** Significantly expressed upregulated and downregulated target genes in JJ012 human chondrosarcoma and control C28 chondrocytes cell lines

| Target genes | Fold upregulation ($P \leq 0.05$; fold changes >2) |
|---|---|
| GAGE2E | 107.88 |
| GAGE2A | 101 |
| GAGE8 | 95.96 |
| GAGE12B | 86.74 |
| GAGE12J | 83.86 |
| GAGE12F | 82.86 |
| GAGE12I | 76.94 |
| GAGE5 | 73.80 |
| GAGE12G | 73.20 |
| LOC645037 | 71.33 |
| GAGE12H | 68.98 |
| GAGE12D | 66.97 |
| MAGEC2 | 60.52 |
| GAGE4 | 59.76 |
| GAGE6 | 59.60 |
| GAGE7 | 55.49 |
| CTAG2 | 47.99 |
| MAGEA1 | 47.68 |
| CT45A4 | 43.46 |
| GAGE12E | 41.51 |
| MAGEB2 | 36.27 |
| SPANXC | 36.07 |
| GTSF1 | 36.00 |
| | Fold downregulation ($P \leq 0.05$; fold changes ≥2) |
| PGM5 | 8.38 |
| INPP1 | 7.29 |
| MX1 | 6.51 |
| KIAA0363 | 6.21 |
| CLDN1 | 6.17 |
| EFEMP1 | 5.83 |
| CKB | 5.56 |
| PDGFRB | 5.36 |
| KIF5C | 5.32 |
| ZSCAN16 | 5.30 |
| ATP9A | 5.17 |
| CCDC102A | 5.16 |
| C6orf150 | 5.12 |
| PJA1 | 5.10 |
| SPARC | 5.08 |
| PALM | 4.97 |
| TMEM42 | 4.92 |
| DBNDD2 | 4.88 |
| RPB4 | 4.88 |
| CACNG6 | 4.76 |
| EFNB3 | 4.74 |
| CHST15 | 4.73 |
| LAMB1 | 4.61 |

reported to decrease glycolysis in the insulin signaling pathway [45], was a downregulated target along with kinesin (KIF5C), known as the gene regulating mitochondrial dynamics. Decreased expression of insulin-like growth factor receptors 1 and 2 (IGF1 and IGF2) was correlated with decreased expression of KIF5C in embryonic stem cells [46]. CTA genes, usually upregulated in normal testis and in certain cancers [47], were upregulated in chondrosarcoma such as GAGE2E (107.88 fold) and GAGE2A (101 fold). They are known to correlate with IGF in Prader–Willi syndrome (PWS) (PWS is typically assumed to be associated with low IGF1 and low MAGE family proteins) [48]. The scientific evidence now supports the idea that DNA methylation is crucially involved in the dysregulation of miRNAs in cancer. This is in part the result of technological advances that enable more comprehensive analysis of miRNA expression profiles and the epigenome in cancer cells, which has led to the identification of a number of epigenetically regulated miRNAs. As with protein-coding genes, it appears that miRNA genes involved in regulating cancer-related pathways are silenced in association with CpG island hypermethylation. In addition, methylation in CpG island shore regions and DNA hypomethylation also appear to contribute to miRNA dysregulation in cancer [49]. Several studies have provided clear evidence that hypomethylation in tumors result in activation of multiple CTA genes [50–52]. This may explain why miR 1909 (chromosome19), upregulated in the chondrosarcoma cell line, results in upregulation of its target gene MAGEA 1. This gene has been reported to be activated due to hypomethylation in other cell lines [52]. It may also explain why genes miR 767-5p, miR 105, and miR 222, located on the same chromosome (18), were upregulated in our experiments or resulted in upregulation of the CTA target genes (Fig. 2). Although global hypomethylation was found to be a comparatively common event in many tumors, little information exists regarding the target genes that are regulated by hypomethylation. It is very possible that these targets, upregulated in our experiments in chondrosarcoma, are under control of epi-miRNAs, a subgroup of miRNAs that control, directly or indirectly, effectors of the epigenetic machinery such as DNMTs, HDACs, and polycomb genes. This observation casts a new light on the functions of miRNAs as both genetic and epigenetic regulators [53]. The role of epigenetic and other possible molecular events responsible for miRNA dysregulation in tumor tissue that leads to malignant transformation is of paramount importance and requires further investigation. The unique expression profiles of different microRNAs in different types of cancers and at different stages in one cancer type suggest that microRNAs can function as novel biomarkers for disease diagnostics and may present a new strategy for microRNA gene therapy [54]. The advantage of microRNAs relative to mRNA gene signatures is shown by the ability of microRNAs to classify poorly differentiated cancers, while mRNA gene

signatures failed to classify them. MicroRNAs can accurately identify cancer tissue origin [55]. In the clinic, microRNA inactivation could be accomplished through a delivery of locked nucleic acid (LNA) antisense oligonucleotides against microRNAs [56]. These molecules are more stable and less toxic than other cancer treatments in order to target oncogenic microRNAs. Sarcomas displaying upregulation of miRNAs, such as those described for RMS and synovial sarcoma, would be candidates for miRNA targeting using synthesized anti-miR oligonucleotides (termed AMOs). AMOs are designed to specifically bind target miRs and sterically inhibit miR association with target mRNAs. Use of AMOs to specifically inhibit miR 21 in human glioblastoma cells and breast carcinoma cells in vitro, and in vivo using mouse xenograft models, resulted in potent inhibition of cell proliferation and reduction in tumor formation. Overexpression of tumor suppressor microRNAs in its turn could be used to treat specific types of cancer. MiRNA levels predict disease prognosis and overall patient survival, and reconstituting their basal levels has been proven to inhibit tumor growth and metastasis [57]. Sarcomas are a rare and heterogeneous group of malignant tumors of mesenchymal origin. Sarcomas such as alveolar soft part sarcoma, leiomyosarcoma, synovial sarcoma, rhabdomyosarcoma, or desmoplastic small round cell tumor have also been described as tumors in which increases in IGF1R levels or some of its ligands have been correlated with sarcomagenesis [58]. Our experimental results very clearly indicated the predominance of dysregulated miRNA expression, related to Notch and IGFR pathways in human chondrosarcoma versus C 28 chondrocytes cell lines. Summarizing the results of the study and the data existing in the literature, we do believe that lost miRNA surveillance of NOTCH and IGFR pathways is involved in and leads to sarcomagenesis. However, the mechanisms and the reasons leading to the deranged surveillance of miRNA as a result of aberrant expression or erratic miRNA processing during malignancy remain to be found.

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**Conflicts of interest** None

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